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import java.io.*;
import java.util.ArrayList;

public class GradeAnalyzer {

    public static void main(String[] args) {

        String filename = "scores.txt";
        ArrayList<Integer> scoresOffTxtDoc = readScores(filename);

        System.out.println("\n");

        scoresOffTxtDoc.add(99);
        scoresOffTxtDoc.add(65);
        scoresOffTxtDoc.add(25);
        scoresOffTxtDoc.add(35);
        scoresOffTxtDoc.add(93);

        int low_value = Integer.MAX_VALUE;
        int high_value = Integer.MIN_VALUE;
        System.out.println("\n");

        for (int x : scoresOffTxtDoc) {
            if (x > high_value) {
                high_value = x;
            }

            if (x < low_value) {
                low_value = x;
            }
        }

        int scoresOffTxtDoc_counterA = 0;
        int scoresOffTxtDoc_counterB = 0;
        int scoresOffTxtDoc_counterC = 0;
        int scoresOffTxtDoc_counterD = 0;
        int scoresOffTxtDoc_counterF = 0;

        double avg = calculateAverage(scoresOffTxtDoc);

        System.out.printf("%s\n", "Average Scores: " + avg);
        System.out.printf("%s\n", "Lowest Score: " + low_value);
        System.out.printf("%s\n", "Highest Score: " + high_value);

        int scoreNumber = 0;
        while (scoreNumber < scoresOffTxtDoc.size()) {
            int grade = scoresOffTxtDoc.get(scoreNumber);

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        if (grade >= 90) {
            scoresOffTxtDoc_counterA = scoresOffTxtDoc_counterA + 1;

        } else if (grade >= 80 && grade <= 89) {
            scoresOffTxtDoc_counterB = scoresOffTxtDoc_counterB + 1;

        } else if (grade >= 70 && grade <= 79) {
            scoresOffTxtDoc_counterC = scoresOffTxtDoc_counterC + 1;

        } else if (grade >= 60 && grade <= 69) {
            scoresOffTxtDoc_counterD = scoresOffTxtDoc_counterD + 1;

        } else if (grade <= 60) {
            scoresOffTxtDoc_counterF = scoresOffTxtDoc_counterF + 1;

        }
        scoreNumber = scoreNumber + 1;

        String outputFile = "report.txt";

        writeReport(scoresOffTxtDoc, avg, high_value, low_value,
scoresOffTxtDoc_counterA, scoresOffTxtDoc_counterB,
            scoresOffTxtDoc_counterC, scoresOffTxtDoc_counterD,
scoresOffTxtDoc_counterF, outputFile);
    }

    System.out.println("\n");
    System.out.println("Grade Distribution:");
    System.out.printf("%s\n", "A (90 - 100): " + scoresOffTxtDoc_counterA);
    System.out.printf("%s\n", "B (80 - 89): " + scoresOffTxtDoc_counterB);
    System.out.printf("%s\n", "C (70 - 79): " + scoresOffTxtDoc_counterC);
    System.out.printf("%s\n", "D (60 - 69): " + scoresOffTxtDoc_counterD);
    System.out.printf("%s\n", "F (below 60): " + scoresOffTxtDoc_counterF);

}

public static ArrayList<Integer> readScores(String filename) {

    ArrayList<Integer> scoreX = new ArrayList<Integer>();

    try (BufferedReader reader = new BufferedReader(new FileReader(filename))) {
        String line;
        int badlines = 0;
        while ((line = reader.readLine()) != null) {
            line = line.trim();

            try {

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        if (line.isEmpty() == false) {
            int numberValueScore = Integer.parseInt(line);
            scoreX.add(numberValueScore);
        }

    } catch (NumberFormatException e) {
        badlines = badlines + 1;
        System.out.println("
##### ");
        System.out.println(" ***** Number format error on parse for int:
*****" + e.getMessage());
        System.out.println("
##### ");
        System.out.println("\n\n");
        continue;
    }
}

System.out.println("=== Grade Analysis Report ===");
int scoresOffTxtDocProcessed = scoreX.size();
System.out.printf("%s\n", "Total scores processed: " +
scoresOffTxtDocProcessed);
System.out.printf("%s\n", "Invalid lines skipped: " + badlines);

} catch (IOException e) {
    System.out.println("
##### ");
    System.out.println(" # Code blows on the buffered reader: # " +
e.getMessage());
    System.out.println("
##### ");
}
return scoreX;
}

public static double calculateAverage(ArrayList<Integer> scores) {

    if (scores.isEmpty()) {
        return 0.0;
    }

    int total = 0;

    for (int score : scores) {
        total = total + score;
    }

    double avg = (double) total / scores.size();

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        return avg;
    }

    public static void writeReport(ArrayList<Integer> scores, double avg, int high, int
low, int counterA,
        int counterB,
        int counterC,
        int counterD,
        int counterF,
        String outputFile) {

        try (BufferedWriter writer = new BufferedWriter(new FileWriter(outputFile))) {

            writer.write("Highest score: " + high);
            writer.write("Lowest score: " + low);

            writer.write("Grade Distribution:");
            writer.write("A (90 - 100): " + counterA);
            writer.write("B (80 - 89): " + counterB);
            writer.write("C (70 - 79): " + counterC);
            writer.write("D (60 - 69): " + counterD);
            writer.write("F (below 60): " + counterF);
        } catch (IOException e) {
            System.out.println("Error writing report: " + e.getMessage());
        }

    }
}

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